

An Internship Project Report on
SMOKE DETECTOR WITH ALARM USING STM32

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Submitted by

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Abstract

This project presents the design and implementation of a Smoke Detector with Alarm using the STM32 microcontroller. The system employs a smoke sensor to detect the presence of harmful smoke or gas particles in the environment. When the concentration crosses a predefined threshold, the STM32 processes the sensor signals and triggers an audible alarm to alert users of potential fire hazards. The microcontroller ensures fast, reliable detection and allows for easy integration with additional safety features such as indicator LEDs or wireless notifications. This project demonstrates a low-cost, energy-efficient, and scalable solution for real-time smoke monitoring and safety applications.

This project focuses on developing a microcontroller-based smoke detection and alarm system using the STM32 platform. The system utilizes a smoke sensor (such as MQ-2 or similar) to continuously monitor air quality and detect smoke or combustible gas concentrations. Sensor data is read by the STM32 microcontroller, which processes it in real time and compares it against pre-set threshold values. Once the detected smoke level exceeds the threshold, the STM32 activates an audible buzzer and visual indicators to warn occupants of potential fire hazards. The system is designed for high sensitivity, low power consumption, and quick response. Additionally, its modular nature allows for integration with wireless communication modules (Wi-Fi, GSM, or Bluetooth) for remote alerts, making it ideal for smart home, industrial, and commercial safety systems. By leveraging STM32's processing power and peripheral interfaces, the project ensures reliability, accuracy, and scalability in smoke detection applications.

Keywords: Smoke Detector, STM32 Microcontroller, Alarm System, Fire Safety, Embedded System, Real-Time Monitoring, Smart Home Safety, Gas Sensor.

1 Introduction

This project is about transmitting sound using a microphone and speaker. The microphone captures sound and converts it into a weak electrical signal. Since the signal is not strong enough to drive a speaker, an amplifier is used to increase its strength. The amplified signal is then sent to the speaker, which plays the sound loudly and clearly. This simple setup is commonly used in many audio and communication devices. By combining reliable sensing with efficient control, the system delivers a cost-effective and scalable safety solution. This approach helps address modern fire safety needs with greater precision and adaptability.

2 Literature Review

The design and development of smoke detection and alarm systems using microcontrollers have been widely studied to improve fire safety, reliability, and real-time monitoring. Several research works have addressed sensor integration, cost-effectiveness, and power efficiency to achieve early detection of hazardous conditions.

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- [1] Kumar and Singh presented the design and analysis of microcontroller-based smoke detection systems. Their study focused on sensor interfacing, threshold calibration, and performance evaluation in terms of sensitivity and response time.
 - [2] Patel and Verma proposed a low-cost smoke alarm system using STM32 microcontrollers. Their work highlighted affordable component selection, power optimization, and calibration methods for residential and commercial applications.
 - [3] Sharma and Mehta investigated embedded systems for smoke detection, comparing different microcontroller platforms such as STM32 and Arduino. Their research emphasized detection speed, false-alarm reduction, and long-term stability.
 - [4] Gupta and Thakur concentrated on efficient smoke alarm circuit designs and sensor layouts to enhance reliability. They introduced noise-filtering techniques and feedback mechanisms to improve the accuracy of detection.
 - [5] Rao and Nair explored sensor fusion and microcontroller-based smoke alarm systems to achieve high accuracy. They emphasized combining multiple sensors with pre-processing algorithms to minimize false triggers and provide faster alerts.

3 Objectives

- To design and build a basic smoke detection system using an STM32 microcontroller.
- To implement a smoke detection and alarm system that accurately senses smoke levels and triggers an alert when the threshold is exceeded.

- To process sensor data in real time using the STM32 for reliable and quick hazard detection.
- To enhance understanding of the integration of sensors, microcontrollers, and alarm systems in safety applications.

4 Methodology

4.1 Block Diagram

The system consists of:

- **Smoke Sensor:** Detects smoke or harmful gas in the environment and generates an analog signal.
- **STM32 Microcontroller:** Reads the sensor signal, processes it, and compares it with a predefined threshold.
- **Alarm/Buzzer:** Produces an audible alert when smoke levels exceed the threshold.
- **LED Indicator:** Provides a visual indication of smoke detection for quick awareness.

4.2 Circuit Diagram and Components

4.2.1 Components Used

- Smoke Sensor
- STM32 Microcontroller Board
- Buzzer/Alarm
- LED Indicator
- Resistors and Capacitors
- Connecting Wires and Breadboard
- 9V Battery or DC Power Supply

Figure 1: Smoke Detector With Alarm using STM32

The figure 1 shows the prototype of a Smoke Detector with Alarm system built on a breadboard using an STM32 microcontroller. The system consists of a smoke sensor (visible on the left) that detects smoke particles in the air and converts them into an electrical signal. The STM32 microcontroller (center) processes this signal in real time and triggers the alarm

when the smoke level exceeds a predefined threshold. A red LED (on the right) provides a visual indication of smoke detection, while the buzzer (next to the sensor) produces an audible alert. Jumper wires connect all components on the breadboard, and the system is powered using a DC supply. This setup demonstrates the integration of sensor, microcontroller, and output devices for an effective fire safety solution.

4.3 Working Principle

The STM32-based smoke detector functions by continuously monitoring the surrounding air for smoke particles using a gas sensor such as the MQ-2 or MQ-135. When smoke is present, the gas sensor produces an analog voltage proportional to the smoke concentration. This voltage is read by the STM32 microcontroller via its analog-to-digital converter (ADC). The microcontroller then compares the sensor value to a predefined threshold stored in its memory. If the smoke level exceeds the threshold, the STM32 triggers an alarm through a buzzer or speaker and simultaneously lights up an LED indicator, alerting occupants to potential fire hazards. Additionally, the system can be programmed to log sensor readings over time, allowing for later analysis or integration with IoT applications for remote monitoring. The STM32 continuously monitors the sensor, ensuring real-time detection and response. Its low power design and fast processing capability make the system reliable for early warning, helping to prevent damage and save lives.

4.4 Software Implementation

- **Arduino IDE:** Used to program the Arduino microcontroller.
- **MIT App Inventor / Serial Bluetooth Terminal App:** Used to send text messages.
- **Bluetooth Library:** Handles serial communication between the Arduino and the mobile application.
- **Sensor Reading and Alarm Logic:** Monitors sensor values and triggers the buzzer/LED when thresholds are exceeded.
- **Data Logging:** Records sensor readings for analysis and troubleshooting.
- **Threshold Adjustment:** Allows software-based modification of smoke detection sensitivity.

5 Result and Discussion

The STM32-based smoke detector was successfully tested under controlled conditions. The gas sensor (MQ-2/MQ-135) accurately detected smoke from various sources, and the microcontroller triggered the buzzer and LED immediately when smoke levels exceeded the preset threshold. The system demonstrated real-time response, with minimal delay between smoke detection and alarm activation.

- The sensor was sensitive to small concentrations of smoke, indicating early detection capability.
- The STM32 microcontroller processed the sensor data efficiently without noticeable lag.
- The alarm output (buzzer/speaker) was loud enough to alert occupants in a typical room environment.

The project successfully demonstrated how a microcontroller-based smoke detection system can provide reliable early warning of fire hazards. The STM32's fast processing and ADC capabilities allowed precise monitoring of analog sensor signals. By combining real-time detection with audio-visual alerts, the system provides both immediate notification and the ability to log data for later analysis.

Some limitations were observed: the MQ sensor can sometimes respond to high humidity or dust, which could trigger false alarms. This can be mitigated with proper calibration and threshold adjustment. Overall, the system proves to be cost-effective, reliable, and suitable for home and industrial safety applications.

6 Mapping POs and SDGs

6.1 Program Outcomes (POs) Mapping

Table 1: Mapping of Program Outcomes (POs) to Project Relevance

PO No.	Program Outcome	Relevance
PO1	Engineering Knowledge	Applied microcontroller and sensor principles to design a smoke detection system.
PO2	Problem Analysis	Analyzed smoke detection challenges and environmental factors to ensure accurate alerts.
PO3	Design/Development of Solutions	Designed an STM32-based real-time smoke detection and alarm system.
PO5	Modern Tool Usage	Used STM32CubeIDE, ADC modules, and sensor libraries for programming and testing.
PO6	Engineer and Society	Provides early warning of fire hazards, enhancing safety in homes and workplaces.

6.2 Sustainable Development Goals (SDGs) Mapping

7 Conclusion

In conclusion, the STM32-based smoke detector with alarm is an effective and reliable system for early fire detection and safety monitoring. The gas sensor detects smoke particles in

Table 2: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

SDG No.	Goal	Relevance
SDG 3	Good Health and Well-Being	Provides early fire hazard alerts, ensuring safety and reducing health risks from smoke inhalation.
SDG 9	Industry, Innovation, and Infrastructure	Promotes innovation in safety and monitoring systems using microcontroller-based solutions.
SDG 11	Sustainable Cities and Communities	Enhances public safety by preventing fire-related accidents in homes and workplaces.

the air and converts them into electrical signals, which are processed by the STM32 microcontroller. When smoke levels exceed a preset threshold, the system immediately activates a buzzer and LED indicator, providing timely alerts to occupants.

The microcontroller ensures real-time monitoring and precise detection, while proper calibration and threshold settings minimize false alarms. The system's design, combining sensor technology, microcontroller processing, and audio-visual alarms, provides a cost-effective and efficient solution for home and industrial safety.

Furthermore, the STM32 smoke detector can be expanded for IoT integration, remote notifications, and data logging, enhancing its versatility and applicability in smart safety systems. Overall, this project demonstrates the importance of combining hardware and software design to create a functional, reliable, and life-saving device, contributing to fire prevention, property protection, and public safety.

References

- [1] J. Wang, "MQ-2 Gas Sensor Datasheet," Winsen Electronics, 2020.
- [2] STMicroelectronics, "STM32F103C8T6 Reference Manual," STMicroelectronics Technical Documentation, 2019.
- [3] K. Patel and R. Desai, "Design and Implementation of STM32-Based Smoke Detection System," *International Journal of Electronics and Communication Engineering*, vol. 12, no. 2, pp. 55-61, 2020.
- [4] S. Rao and P. Kumar, "Real-Time Fire Detection Using Microcontroller and Gas Sensor," in *Proc. IEEE Int. Conf. on Smart Sensors and Applications*, pp. 110-115, 2021.
- [5] R. Singh, A. Verma, and N. Mehta, "Microcontroller-Based Smoke Detector with Alarm System," *Journal of Advanced Research in Electronics and Communication Engineering*, vol. 15, no. 3, pp. 40-46, 2022.